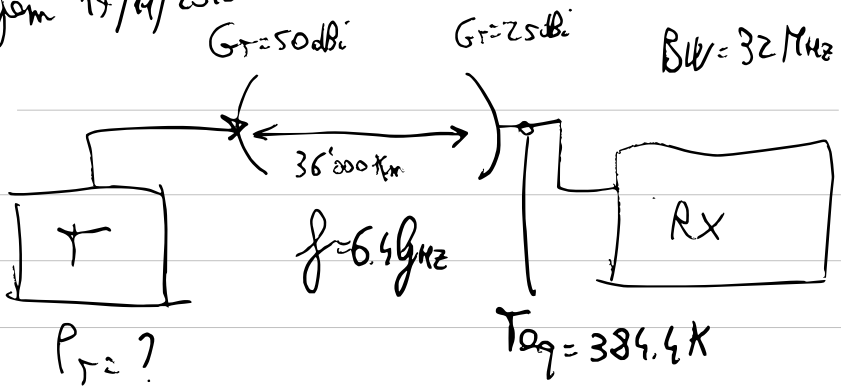
fam 19/4/2015



① COMPUTE TX POWER for SNR = 30dB

$P_n = KT_{eq}B = 0.16975\,pW = -127.7\,dBW = -97.7\,dBm$

$\Rightarrow P_{RX} = -67.7\,dBm \quad (P_n + 30dB)$

$P_{RX} = P_{TX} + G_{TX} - FSL + G_{RX}$

$FSL = 20\log\frac{4\pi d}{\lambda} = 199.7\,dB$

$\Rightarrow P_{TX} = P_{RX} - G_{TX} - G_{RX} + FSL = 57\,dBm = 27dBW$

② If P_{TX} is a 2-TONE SIGNAL, $\Delta_P = 10.6dB$
what is the P_{1dB} of the TX to have $P_{INTERMOD} = P_n$ AT the RECEIVER?
(NEGLECTING COMPRESSION)

$P_{INT} + G_{TX} - FSL + G_{RX} = -97.7\,dBm$

$\Rightarrow P_{INT} = 27dBm \quad (C/I = SNR)$

$P_{INT} = 3P_{TX} - 2IP_3 - 6dB \Rightarrow IP_3 = \frac{P_{INT} + 6dB - 3P_{TX}}{-2} = 69\,dBm$

$P_{1dB} = IP_3 - \Delta P = 58.4dBm \Rightarrow BO = 1.4dB \Rightarrow$ VERY LOW (LINEAR THE MAIN TONES ARE ALSO COMPRESSED)